

A Review on Moving Target Defenses and Cyber Deception Technology with Software Defined Network in Cloud Computing

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Abstract—The increment of the web platform and end-user throughout the last few years demands the growth of security platforms while boosting the performance. The current systems can easily cope with the known threats, but it also allows room for unknown threats, ultimately making the system more vulnerable over time. Most of the detection systems are entirely passive, leading to a failure to comprehend unknown hazards and prevention systems that allow the opportunity to carry out a different approach without any difficulty. Software-Defined Networking (SDN) has emerged as a framework for centralized command and control in cloud data centric environments. SDN separates data and control plane, which provides network administrator better visibility and policy enforcement capability compared to traditional networks. The SDN controller can assess reachability information of all the hosts in a network. There are many critical assets in a network which can be compromised by a malicious attacker through a multistage attack. Thus we make use of centralized controller to assess the security state of the entire network and pro-actively perform attack analysis and countermeasure selection. Moreover, our framework has a comprehensive conflict detection and resolution module that ensures that no two flow rules in a distributed SDN-based cloud environment have conflicts at any layer; thereby assuring consistent conflict-free policy implementation and preventing information leakage.

Index Terms—Cyber Security, Cloud Computing, Moving Target Defense, Cyber Deception, SDN, NFV

I. INTRODUCTION

As the web platform increases alongside end-users, the considerable load of traffics needs to be in checking. After checking the solution distribution, it is concluded the intrusion detection system works efficiently than most of the other

solution which can be provided. It is suggested that IoT attacks have been increased by 150% in only the first half of 2019 compared to 2018. According to the famous Asian newspapers, in this COVID-19 pandemic situation, the number of cyber incidents rose by 200% over the first two months. This sort of situation indicates that cyber threats will drastically increase over time. The recent attack in Beijing on July 28, 2020, where Hackers linked to the Chinese government have infiltrated Vatican computer networks. One of them is the Roman Catholic Church's Hong Kong-based representative, a U.S. firm that tracks state-backed cyber attacks.

Cloud applications are becoming pervasive more due to scalability ,accessibility. Businesses and organizations are shifting their workloads across various IaaS platforms, including Amazon Web Services (AWS), Google Cloud, and Microsoft Azure. As a consequence, hybrid cloud computing that incorporates local servers with private and third-party public cloud services is one of the latest developments in cloud computing. Among all the recent technologies in cloud computing, application containers have become very common in the recent past as they provide ease in maintaining and even transferring application code. After its introduction, Kubernetes has been the dominant container platform and is expected to be one of the best developments in cloud computing future technology [1]. Over the last year, more organizations have moved their maximum dependency on cloud computing, storage and technology. However, organizational problems and difficulties still remain unsolved [2]. While the future of cloud computing technology is

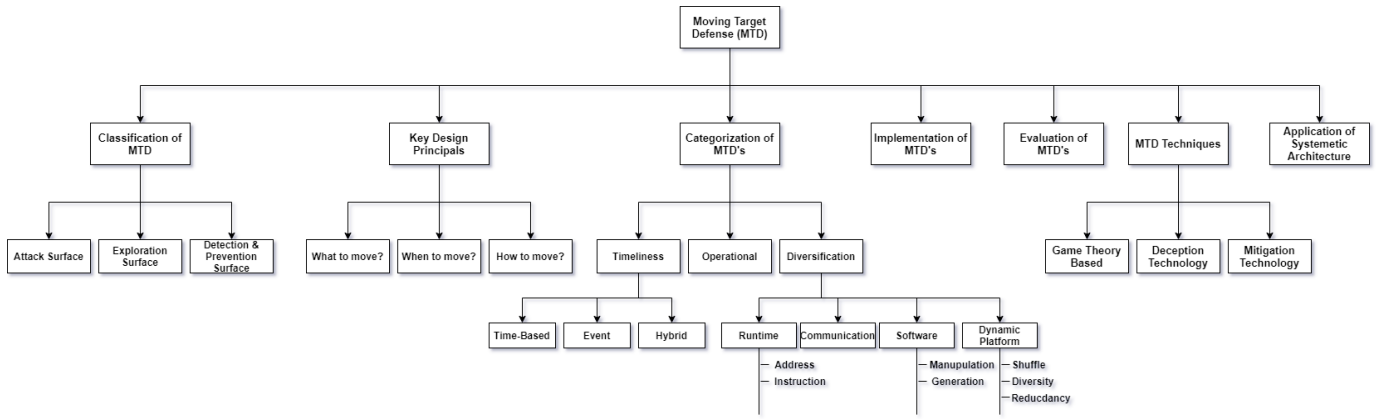


Fig. 1. Taxonomy of Moving Target Defense Survey

quite bright for the global economy. Due to this growing intake of cloud technologies, attacks are also looming, and we can see that every useful technology that is adopted by any company will be the primary focus of attackers [3]. According to the CSA Releases New Research -Data Breaches, Misconfiguration and Inadequate Change Control, Lack of Cloud Security Architecture and Strategy, Insufficient Identity, Credential, Access and Key Management, Account Hijacking are some of the egregious threat that are being critical concerns.

Mobile device attacks are a new emerging risk or upgrades are not produced for known vulnerabilities it therefore also increases security-related risks. Organizations should address the risk of unsuccessful, untested or unauthorized changes being implemented that could. Cloud computing in information risk management is the ability to manage risk more effectively. There are many new technologies emerging at a rapid rate, each with technological advancements. Due to the cloud computing technology application in the rail system, the traditional applications of the railway information on the issue of compliance, basically from the point of view about technical to consider the cloud computing security problems, but this is not enough. It's because of this reason traditional security measures like identification, authentication and authorization is not enough in case of cloud computing.

Although cloud infrastructures have increased the automation options for managing IT systems, the introduction of making a system change proactively as a means to eliminating the asymmetric advantage the attacker has on time, known as Moving Target Defense (MTD) techniques. This paper represented the MTD platform for Cloud-Based IT Systems (MTD CBITS) and evaluated its practicality [4]. Moving Target Defense (MTD) is a technique that increases uncertainty due to an ever-changing attack surface. In combination with Software Defined Networking (SDN), MTD has the potential to reduce the effects of a large-scale

cyber attack. This paper evaluated the defense techniques of moving-target using Software Defined Networking and investigated their effectiveness [5]. However, the replication approach taken by cloud computing to provide resiliency leads to an increase in the number of ways an attacker can exploit or penetrate the systems. This calls for designing cloud systems that can accurately detect anomalies and dynamically adapt themselves to keep performing mission-critical functions even under attacks and failures. This paper represented a self-adaptive resiliency approach for cloud enterprise systems that employs a live monitoring and moving target defense based approach to automatically detect deviations from normal behavior and reconfigure critical cloud processes through software-defined networking to mitigate attacks and reduce system downtime [6]. In most cases due to lack of sophisticated security frameworks and trained personnel, such campus private clouds (CPC) are not fully equipped to handle sophisticated integrity, availability, and confidentiality attacks. This paper demonstrated the utility of a cost-effective, and implementation-ally simpler Moving Target Defense (MTD) based cloud resource adaptation approach that significantly reduces the probability of attack success [7].

Software-Defined Networking (SDN) is an approach to network management which separates data and control plane and provides network administrator a better visibility and a better policy enforcement capability than the traditional networks. It is emerged as a framework for centralized command and control in cloud data centric environments. Network Function Virtualization (NFV) is a concept where IT virtualization technologies are used to virtualize entire classes of network node functions into building blocks which are connected or can be chained together to create communication services. NFV has emerged to comprehensively address several networking issues and provide the mechanisms that help locating the source and types of attacks as a reactive defense mechanism.

II. BACKGROUND & RELATED WORK

In this section, we are going to briefly explain the concepts that are related to MTD and Cloud alongside the threat each intervenes based on the paper surveyed. The work on MTD and their techniques are growing and in recent years number of surveys and researches have been increased almost double compare to last decade. Depending on the survey we have created an outline which follows-

A. Advance Persistent Threat

Advanced Persistent Threats (APTs) refers to a distinct set of attacks against a high-value target organization that differs from normal cyber attacks in several ways. First, APTs are achieved by a group of highly-skilled attackers who are well-funded. According to Mandiant Report, APTs such as Operation Aurora, Shady Rat, and Red October have been used on a global scale for industrial espionage. Oftentimes, the attackers mounting APT work closely with government organizations. Second, an APT attacker is extremely persistent; they pursue the objectives repeatedly, often over an extended period, can adapt to a defender's efforts in trying to resist the attack, and determined to maintain a level of access within the defender's system required to achieve their objectives. Third, the key objective of APTs is to either exfiltrate information or undermining the key services in the network using multiple attack vectors.

B. Stages of APT

An APT can be broken down into five phases— reconnaissance, foothold establishment, lateral movement, data exfiltration, and post exfiltration. These can be mapped to different phases of the Cyber Kill Chain. The reconnaissance phase in APTs maps directly to the reconnaissance phase in CKCs, which was described earlier. The foothold establishment phase comprises of the weaponization and the delivery phases of CKC. The attackers use information gathered from the reconnaissance phase in order to prepare an attack plan and eventually exploit vulnerabilities uncovered in the target organization's web application, databases, and other software.

1) *APT Defense Techniques:* In this section, we provide a brief overview of the various defense techniques and highlight how each of these defense mechanisms is effective in various parts of a Cyber Kill Chain (CKC) in Table II. This discussion will help the reader better understand some of the MTD techniques that use these traditional defenses as a building block.

2) *APT Installation:* Once the attacker has gained a foothold into the target environment, they can try to move laterally in the target environment and gain access to other sensitive hosts and critical information in the organizational network in the lateral movement phase of APT. In this setting, the attacker needs to continuously explore (perform reconnaissance) and exploit the various components of the defender's system, mapping to the exploitation phase of the CKC. In the data ex-filtration phase of APT, the attacker tries to exfiltrate

the collected data to their command and control (C&C) center. Most of the intrusion detection and prevention systems do ingress filtering and not egress filtering, thus, data exfiltration can often go undetected. The goal of the attacker in post ex-filtration phase is to maintain persistence in the system for a long duration of time until the funding for attack campaign is finished or the goals of attacking the organization/government are fulfilled.

C. Distinctive Security Concepts in Cloud

A huge amount of security-related information is present on the web— in social engineering websites, phishing sites, and dark-web forums— including discussion about a particular target product or company. As discussed by Glass et. al., the problem of exploring and analyzing the web for information should provide Security relevant information discovery capabilities Situation awareness by performing real-time inference over the available information, and predictive analysis to provide early warning for any likely security attacks.

There are two types of IDS agents that can be deployed in the network under attack, i.e., Network-based IDS (NIDS) and Host-based IDS (HIDS). NIDS such as Bro and Snort, use signature match techniques based on known attack patterns and can flag incoming network traffic as malicious or benign. HIDS such as auditd or tripwire, on the other hand check the Indicators of Compromise (IOCs) on the physical server or VM, in order to identify malicious activity, e.g., privilege escalation attempt by normal user.

A security mechanism, which can be used to detect, deceive or in some cases counter a malicious user trying to gain access to key network services. Honeypots such as Cowrie can help in better understanding the attacker's tools and tactics. The connection requests from the attacker are directed to a decoy service, which mimics the behavior of the normal service and logs the attacker's activity.

Although there exists a large set of defense mechanisms, attackers often use clever techniques to evade detection or prevention. SANS discusses methods like obfuscation, fragmentation, encryption and Denial of Service (DoS) attacks against signature-based detection tools. Detection based on HIDS can also be evaded by a skilled attacker using techniques such as file location manipulation (using directories or files white-listed by IDS), application hijacking, etc. Furthermore, machine learning models that can help overcome some of the shortcomings of signature-based detection tools can themselves be fooled by adversarial attacks. On similar lines, deception techniques such as DNS redirect and Honeypot can help in deceiving the attacker temporarily, but over a longer period of time, the attacker will eventually change their attack methodologies thereby reducing the effectiveness of these defenses. Stojanovski et. al. performed experimental analysis on how to bypass DEP protection on Windows XP systems. Thus, there is a need to perform intelligent manipulation to assure the attacker's likelihood of reaching the desired goal

TABLE I
COMPARISON OF CONCEPTS AND TECHNIQUES OF MTD, COMBINATION OF MTD & CYBER DECEPTION TECHNOLOGY IN CLOUD DOMAIN AND THEIR EFFECTIVENESS BETWEEN OUR SURVEY AND OTHER SURVEY PAPERS

Review Study	Application Domain	Key Design Principals	APT	Implementation	Design Techniques	Evaluation	Research Direction
[8]	Cloud,SDN	✓	✓	✓	✗	✗	✓
[9]	Unclassified	✓	✗	✗	✗	✗	✗
[10]	Cloud,SDN,Iot	✓	✗	✓	✗	✗	✓
[11]	Cloud,SDN,Iot	✗	✗	✗	✓	✗	✓
[12]	Unclassified	✓	✗	✗	✗	✗	✗
[13]	Cloud,SDN,Iot	✗	✗	✗	✓	✗	✓
[14]	Unclassified	✓	✗	✗	✓	✗	✓
[15]	Cloud,SDN,Iot, CPS,Enterprise	✓	✗	✗	✓	✓	✓
Our Survey	Cloud,SDN,NFV, CPS,IoT,Edge,Fog, Enterprise	✓	✓	✓	✓	✓	✓

is limited, even if they can evade the traditional detection methods. Additionally, the defense mechanisms discussed in the Section II-D target known attacks often with easy to detect signature patterns.

D. Enabling Technologies

1) *Software Defined Network:* Software-Defined Networking (SDN) is an approach to network management which separates data and control plane and provides network administrator a better visibility and a better policy enforcement capability than the traditional networks. It is emerged as a framework for centralized command and control in cloud data centric environments.

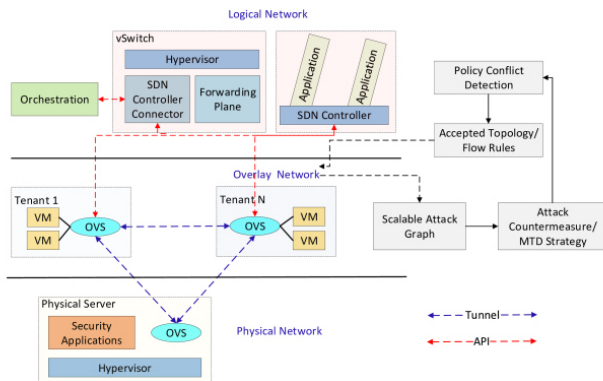


Fig. 2. SDN Based Logical Network

2) *Network function virtualization*: Network Function Virtualization (NFV) is a concept where IT virtualization technologies are used to virtualize entire classes of network node functions into building blocks which are connected or can be chained together to create communication services. NFV has emerged to comprehensively address several networking issues and provide the mechanisms that help locating the source and types of attacks as a reactive defense mechanism.

III. KEY DESIGN PRINCIPLES FOR MTD IN CLOUD

This section discusses different key design that are used and talked about in recent years. Cheng Lei et al.[1] proposed three

basic models of MTD based on strategy formulation, transformation mechanism, and effectiveness evaluation. According to Cheng Lei et al.[1] MTD strategy formulation means producing the optimal defense strategy by examining and simulating existing network threats. They divided this in three main categories which are Game Theory, Machine Learning Control Theory, Evolution Theory. Another technique used in MTD based cloud computing is shuffle and diversity. Hooman Alavizadeh et al.[2] MTD techniques based on shuffle and diversity in a single MTD and evaluated its effect on cloud environment.

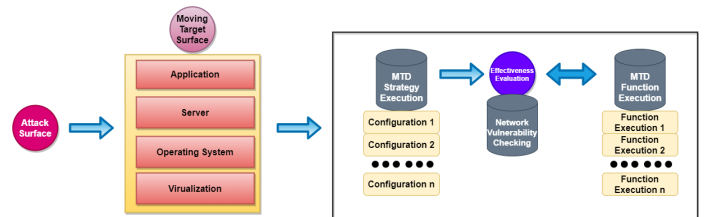


Fig. 3. Attack Surfaces Methods of Proactive Moving Target Defense

Game Theory based MTD : Game Theory is a simulative model between self-interested agents. It studies the decision making process and achieves maximum benefit for an agent. In cloud computing and MTD based environment game theory suggests a number of strategies according to the defender and attacker. Cheng Lei et al.[1] divided game theory models in two parts which are: strategy under complete information, strategy under incomplete information. In strategy under complete information they included two player stochastic dynamic game model and markov decision model. On the other hand strategy under incomplete information, they included zero-sum game model, leader-follower game model (defender as leader and attacker as follower) and bayesian stackelberg model. According to Yan Sun et al.[3] process of attack and defense contains features of objective opposition, strategic dependence and cooperative relation. They divided the MTD based game theory models into four parts which are: classical game model, markov game model, differential game model and evolutionary game model. These models are applied and working models

TABLE II
A SUMMARY OF LITERATURE REVIEW BASED OF IMPLEMENTATION TECHNIQUE AND NETWORK LAYER

Type	Application Layer	Key Techniques	SDN/NFV	APT	Metrics	Research Level	References
Shuffling	Network	Simpv	✓		CIA Metrics	simulation	[16]
Redundancy	Network	Cloud Lab		✓	Attack Tree	simulation	[17]
Diversity	Network	Smart Grid	✓	✓	Attack Tree	emulation	[18]
Shuffling	Network	CMS			CIA Metrics	simulation	[19]
Shuffling	Network	LLVM	✓		Attack Tree	simulation	[20]
Diversity	Network	Matlab			Attack Tree	emulation	[4]
Shuffling	Abstract	Geni			QoS Metrics	emulation	[21]
Redundancy	Application	ArcSync			Cost Metrics	simulation	[22]
Diversity	Network and Application	Geni		✓	CIA Metrics	commercial	[23]
Hybrid	Network and Application	RPAF			Risk Metrics	simulation	[24]
Hybrid	Network and Application	Core			QoS Metrics	emulation	[25]
Shuffling	Network	RocketFuel			Attack Tree	emulation	[26]
Shuffling	Network	Customized			Policy Conflict	commercial	[27]
Diversity	Abstract	Unidentified			Risk Metrics	both	[28]
Shuffling	Network	Gamebed			Cost Metrics	simulation	[29]
Diversity	Application	Nox			Attack Tree	commercial	[30]
Redundancy	Network and Application	Customized		✓	CIA Metrics	simulation	[31]
Redundancy	Network and Application	PlanetLab			QoS Metrics	emulation	[32]
Shuffling	Network and Application	Unidentified	✓		Policy Conflict	emulation	[33]
Shuffling	Network	Personalized			QoS Metrics	commercial	[34]
Hybrid	Network	Geni			Risk Metrics	simulation	[35]
Hybrid	Application	Onos			Attack Tree	simulation	[36]
Diversity	Network and Application	Customized		✓	CIA Metrics	simulation	[37]
Hybrid	Network	Polyverse			QoS Metrics	simulation	[38]
Redundancy	Network	Deception Grid			Cost Metrics	simulation	[39]
Hybrid	Network	Open Daylight Helium			Risk Metrics	emulation	[40]
Redundancy	Network and Application	Customized			Attack Tree	commercial	[41]
Redundancy	Application	Mininet			Policy Conflict	simulation	[42]
Hybrid	Network	Unidentified			Risk Metrics	emulation	[43]
Diversity	Network	POX			Policy Conflict	both	[44]
Diversity	Network	Unidentified			Cost Metrics	emulation	[45]
Hybrid	Network and Application	RyuSDN	✓		CIA Metrics	emulation	[46]
Diversity	Network and Application	SLAAC			Attack Tree	simulation	[47]
Diversity	Abstract	MUTE			QoS Metrics	both	[48]
Shuffling	Network and Application	KDD			QoS Metrics	commercial	[49]
Shuffling	Network	Customized			Attack Tree	commercial	[50]
Shuffling	Network and Application	Unidentified		✓	Policy Conflict	analytic	[51]
Redundancy	Network	Morphisec			Cost Metrics	emulation	[52]
Redundancy	Network	Customized	✓		CIA Metrics	analytic	[53]
Diversity	Network	Customized	✓		Risk Metrics	emulation	[54]

for MTD in game theory. The classical game model suitable for short duration attack environment, the markov model is suitable for repeated attack system, differential model is suitable for irrational environment, evolutionary game is suitable for sides finding optimality. In conclusion MTD based game theory model is a strong security model though it has simulative characteristics that needs time for the specified system.

Shuffle and Diversity based MTD : Hooman Alavizadeh et al.[2] implemented shuffle technique utilizing a VM-LM method. This technique rearranges system configuration. On the other hand diversity technique replaces the system components but keeps a normal running system. And then they implemented both which share a minimal deployment that keeps changing the system and makes the cloud environment unpredictable from attackers. Although it's an effective MTD technique which strengthens the security, the techniques must be combined. And if these techniques is not deployed properly it harms the real system. Diversity technique can

enhance AC(attack cost) and RoA(return on attack) based on the attacker's perspective. Shuffle technique shows security metrics like R(Risk) and ASP(Attack success probability) which mainly capture the security from cloud providers' point of view. Shuffle and diversity can be used without others to develop security on the cloud network.

Machine learning based MTD : The cloud computing data are mostly efficiently used t=in machine learning based MTD. The data is clustered and then control theory is used with corrective function that provides the correct system environment. Therefore any types of informality in the system is seen and worked through. There are different methods or approaches to implement machine learning. One approach is to calculate triggering time of any informality and based on the cluster the machine learning algorithm changes system direction. Open-loop with closed-loop defense was proposed by Adams et al.[4] that makes our system unpredictable by using multi-elements. Zeng et al.[5] proposed an MTD strategy that analyzes defensive environments. Machine learning is

mostly used according to the data produced and analyzed from a cloud system which makes it more vulnerable when the system provides redundant data. Therefore machine learning doesn't cover security issues with the MTD. But based on strong algorithms it can enhance security issues.

Cross-Layer based MTD : This technique transforms multi-elements in different layers and uses deception as a defensive tool from attackers. There are different ongoing patterns of cross-layer techniques. Net Maneuver commander (NMC), Applications participating own defense (APOD), Advanced applied Application (A3) and Trusted dynamic logical heterogeneity (TALENT) are hugely used. In NMC the elements are randomized based on geographical destination and the flexibility is improved. APOD analyzes security threats and formulates its own security policies that enhance for future attacks. A3 adopts isolation containers in a controlled way and then modifies from past security breaches. In the meantime TALENT randomizes operating systems and makes a virtual runtime environment so the system doesn't fall. These are a couple of cross-based MTD that enhance security based on the applications used in the system, past attack characteristics and data values such as destination of physical server, port or hardware. In conclusion cross-layer based MTD is a self adaptive system that randomizes, formulates and improves its own system based on the system and applications used.

These are the MTD techniques that can be used to enhance the security of cloud computing. As the techniques shown above are implemented on the real system, users need to be aware of the techniques when used because most of these can directly harm the system. The system data, port or servers are moved and controlled by the algorithms. Therefore the attackers are unable to predict a move against it.

IV. MTD TAXONOMY IN CLOUD COMPUTING

In this section, we are going to explain how MTD can protect the infrastructure in different layers of protection and surfaces while being able to prevent sacrifices. We can differentiate the MTD into these four types of MTD Shiftings:

A. MTD by exploration surface shifting

These works ensure that the attacker will not be able to gather any information about the model of the system by taking any kinds of actions for the exploration. In this process the attacker will always gather false information from the reconnaissance phase. For example, while scanning for open ports and discovering vulnerabilities, attackers will gather inaccurate info to take steps. Moreover, it can be said that in this process an adversary has to work in a faulty view. Here a concept is proposed. In the proposed work, random virtual IP addresses are assigned to the moving target hosts in an unpredictable and decentralized fashion. Each and every configuration represents a mapping from the virtual IP address to the set of hosts of the particular network.

B. MTD by attack surface shifting

Here invalidation of the action of the attacker is seen. For taking any action, the attacker does so much exploration and

after gathering enough information to take the appropriate action, the attacker moves forward. Attack surface shifting is the process of MTD where an attacker's attack option gets invalid by the system. For example, an attack will be useless if the attacker can do the exploration or take any attack on Linux-based OS but the whole package is on Windows OS. Here a concept of work has proposed. In the work, a full-stack system is divided into several layers to provide the functionality which has been decided before for each layer. For this, some specific technologies are being developed for each layer to do the functionalities which they are supposed to do. but at the point when all the layers come to have a full stack system, In spite of using all the technology of each layer, they just use a very few combination which can fully fill the use case demands.

C. MTD by detection surface shifting

This is a work where cybersecurity is developed in this way that it can detect the attacks by enriching the proper information from the traffic on the wire or from the behavior of a request on a host machine. So the system can take proper steps quickly against the attack. A concept of work has proposed here to thwart known attacks on the system. Here every state of the system is composed of variables, each of the variables indicates if a certain attacking zone or any vulnerability in the system has been exploited or not. And based on the information, decides when and how to move.

D. MTD by prevention surface shifting

In this system, there are some modeling of defenses to make the attacker unsure about the defense procedure that is appropriate on that time, also make the attacker bound to spend more resources for the exploration process.

V. MTD MODELING FOR ATTACKS IN CLOUD AND LIMITATIONS

In this section, we delineate the attack characteristics considered in MTD techniques. To be clear, we elaborate on the features of advanced attacks and attackers. We will illustrate attack types and will also consider common MTD approaches. Lastly, we will discuss MTD modelling for preventing the attack and limitation of those.

A. Advance Attack Characteristics

MTD strategies have been designed to control more intricate, smart and persistent attack cases armed with more advanced resources. The main features of these advanced attackers are focused as below.

1) Persistent Attack: In recent studies, we discovered that the attackers do not stop just staging one attack; they opt for multi-level, multi-stage and persistent attack. We can consider APT attacks in this case. [55]–[58] Multi-stage attacks display the persistent attackers behaviour, and that starts with scanning attacks and carrying out attack delivery and exploitation.

2) *Adaptive Attack*: Attackers adapt to rapidly changing device parameters and external environmental environments, taking into account all physical and cyber accessibility factors. These attackers are intelligent in terms of their capabilities, carrying out adaptive attacks [41], [59], [60] that wisely control their resource constraints. In the meantime, They become opportunistic to penetrate the system to compromise the whole system.

3) *Stealthy attacks*: Attackers do not always display any traceable attacking actions. They carry out attacks in an extremely sleuthy way [61], and camouflage as a good and trusted quality of decent people. However, at a time when the attack is estimated to cause significant injury or damage to the device, the attack activity is revealed. So they're going to keep stealthy until the revelation of attack.

4) *Incentive-driven attacks*: MTD is designed to handle and critically intelligent and savvy attackers. For achieving a maximum result from an attack with minimum cost spent, they are quick to perform an attack to damage any system. Considering this factor, we should assume the attacker to be a reasonable agent susceptible to rewards, such as attack success at a minimal cost [62].

B. Attack Types

1) *Reconnaissance attacks*: Attackers stage reconnaissance attack in order to garner information and intelligence over the target device before the real attack is even initiated. Attackers operate a custom set of software as a tool to find detail oriented attributes of a targeted device such as OS types, IP and Device address, port number, network topology. We can use Nmap [63] as an example. An attacker usually performs attacks by sending probe packets, packet sniffing, and scanning ports. As a countermeasure network address based approaches have been offered [64], [65]

2) *DoS (or DDoS) attacks*: A Vast number of botnets are comprised of zombie machines to launch DDoS attacks by swamping the targeted device by sending off enormous amounts of targeted messages. The messages are launched from various accounts and channels to force devices to initiate shutdown commands automatically or gainsay services to authentic users. DDoS attacks also entail time-based attacks that can result in unwanted skipping of critical deadlines in order to update the system and incur a loss to perform important functions in targeted systems. The use of secret proxies, IP/Ports shuffling have been suggested to circumvent the attacks in these MTD research works [66]. Meier et. al. [67] demonstrated a framework namely NetHide as a network obfuscation approach that can be used to combat future attacks namely LinkFlooding attacks performed by advanced attackers. NetHide is capable of adjust and alter the route tracking probe in the data plan. They have also illustrated a trade-off which NetHide can perform between security and usability minimizing the risk of attack success by 1%. while providing 90% and 72% accuracy and utility.

3) *Buffer overflow attacks*: this type of attack leverages the lack of buffer boundary checking. Attackers manipulate buffer

overflow issues by overwriting an application's memory. This modifies the program's execution path, causes a response that destroys files or reveals confidential data. For example, an attacker can enter extra code and send new commands to an application to obtain access to critical systems. In order to thwart this attack common memory protection techniques such as Address Space Layout Permutation [18] can be introduced.

4) *Worm attacks*: A computer worm is a type of malware that disseminates its malignant copies from device to device. A worm can duplicate itself without any human interaction, and it does not need to append itself to a software program in order to incur destruction. Network Address Space Randomization can be used as a counter step that can generate random host IP addresses [61]. IP shuffling is one of the examples of NASR

5) *APT attacks*: APT attacks are eminent as one of the advanced forms of attacks order to exploit device resources, manipulating them and carrying out exfiltration attacks (e.g. revealing classified or proprietary data to unauthorized parties) by executing stealthy, consistent and proactive attacks. The characteristics of APT attackers can be mapped to the cyber kill chain. APT attacks are also known as multi-phase attacks, multi strategized and persistent attacks.

6) *Side channel attacks on VMs*: Leveraging on the commonly shared CPU cache of VM attackers can have access to sensitive information through side channels attacks [68].

7) *Web based Attack*: SQL injection, False data injection, directory traversal are some of the name of attacks that can disrupt normal activities of web applications. Diversity in software stack and redundancy can circumvent these types of attacks [69], [70].

Limitation of MTD Attack Models: Lack of the consideration of Smart, intelligent attackers Detection of defense patterns is prevalent for attackers who are intelligent by using machine learning algorithms. Shuffling based MTD techniques deal with attacks. So they can easily grasp the type of MTD technique and operation that are being used.

Short amount of scenario MTD approaches have considered in small set of attacks hence, attackers can easily penetrate a system by repeating the strategies

C. Attacking modelling techniques

Although the organizations don't want to leave any stone unturned in order to circumvent any imminent and destructive attack. From using advanced management tools to software patching and analysis of the system records following best practices. Still adversaries developed by skilled and savvy attackers reigns to exploit any vulnerabilities present in the network by using zero day attacks, sophisticated ransomware etc. In order to detect any cyber detect beforehand of any attack, it is quintessential to collect information and thereby deploy intelligent cyber defense mechanisms. Studying and analyzing specific threats from the lens of an attacker is the perfect weapon for the detection and therefore the mitigation of sophisticated attacks in networks, specially Advanced Persistent Threats [71], [72]. In order to find correlation, collection and classification of data linked to a cyber attack. Lockheed Martin

replicated the cyber kill chain [65] structure from the military model – originally structured to define, plan to strike, engage and destroy the target. After its inception, the kill chain has developed to help anticipate and detect insider risks, social engineering, sophisticated ransomware, and disruptive attacks. There, we will first delineate the different stages of the CKC followed by a brief description of APTs and the correlation between APT and CKC. This knowledge will later further us to fathom how MTDs can achieve success against the different phases of an APT. Cyber Kill Chain lets us grasp and battle ransomware, data breaches, and advanced persistent attacks (APTs).

Reconnaissance: This phase can be interchangeably called as the Observation phase. The attacker garners detailed information about the targeted system in this stage. The reconnaissance stage in cyber kill chain shares similarity to the reconnaissance stage in APTs.

Weaponization: In this step, the intruder creates a malware weapon like a virus, worm or such in order to exploit the vulnerabilities of the target.

Delivery: Attacker dispatches malignant payload to the targeted environments (i.e. emails). Weaponization and delivery stages constitute the foothold establishment of the cyber kill chain.

Exploitation: Upon identifying (zero day) vulnerability in the targeted system, the attacker detonates the blistering attack and compromises the whole machine.

Installation: After gaining higher privileges, they install malwares that includes ransomware and remote-access Trojans.

Command and Control (C&C): Installed ransomware makes the use of command and control connections for downloading encryption keys.

Actions on Objectives: Attack goals are achieved through data exfiltration and service disruption.

Upon gaining a foothold into the targeted system, attackers move laterally in order to obtain sensitive hosts and crucial information in the lateral movement. In lateral movement it is difficult to detect if any attack is ongoing. Therefore MTD approaches can fall short. In this stage attackers explore and exploit various triggering components of the system and system compromises, thus it skips the detection. Exploration and exploitation in these settings links to the exploitation stage of cyber kill chain. Attackers endeavour to exfiltrated the gathered data to C&C center in data ex-filtration phase of APT. Attackers maintain persistent attacks for a long duration in this stage.

D. MTD Modeling

In this section we will shed light on key MTD modeling based on popular MTD approaches. Three MTD approaches are 1. Game theory based MTD 2. Genetic Algorithm based MTD 3. Machine Learning algorithm based MTD

1) *Game Theory Based MTD:* MTD techniques are based on manipulating the attack surface which add another layer of protection and add additional cost for launching an attack

TABLE III
DIFFERENT NETWORK TECHNIQUE CAN BE USED IN DIFFERENT PHASES OF CKC TO MITIGATE ATTACK

Phase	Detect	Deny	Disrupt	Degrade	Deceive
Reconnaissance	Web Analytics	X	X	X	X
Weaponization	NIDS	NIPS	X	X	X
Delivery	Vigilant user	Proxy filter	AV	Queuing	X
Exploitation	HIDS	Patch	DEP	X	X
Installation	HIDS	chroot	AV	X	X
C2	NIDS	ACL	NIPS	Tarpit	DNS redirect
Action on Objectives	Audit log	X	X	QoS	Honeypot

causing service unavailability for a significant amount of time. Therefore, game theory based MTD is contrived on the concept of gain and loss. This approach can be thought of as a game between attacker and defender. Defender's goal is to reposition the attack surface to minimize the perill of damage as a defense to the action of the attacker [73]. From the side of an attacker's view, the attacker solely tries to launch an attack without incurring a heavy loss. Here's the defender is MTD-Based defender on the opposite of intelligent attacker.

In this section, we will illustratively discuss the prevalent game theoretic approaches designed on the basis of various MTD techniques. a) General Game Theoretic approach b) Bayesian Stackelberg Game b) Stochastic game

General Game Theoretic approach: The principal characteristics of a general game framework is based on the assumption that attacker and defender are logical and the common goal is to increase their chance to achieve their corresponding success. Although game theory is not considered in this approach, hence gaining the highest utility based game is the best option in case of choosing the most preferable strategy.

Bayesian Stackelberg Game: Bayesian Stackelberg Games proved to be successful in designing MTD based defenders to defend against various cyber security adversaries and cyber attacks.. The game is played among two players. Between them one is a leader and one is a follower. A leader poses an action before a follower. Follower investigates the impact of the action taken by the leader and launches action for payoff. Optimizing the effectiveness of a leader's action is the aim of the action of the follower. **Stochastic Game:** Shapely developed a stochastic game that reflects the changing positing between one or more players on the basis of probabilistic dynamics. There are multiple stages in this game approach where every player can start from some states. Each player has independence to opt for an action and receive an effect with respect to that particular action and the current state. Repetition of the dynamics happens when each player starts from a new state and the new states are based on probability based transitions. Two player stochastic game has been designed by Manadhata[30] in order to determine paradigmatic MTD strategy based on the diverse attack surface. The author determined defense strategy by modeling the attacker's profile

and by using subgame perfect equilibrium concept.

2) *Genetic Algorithm-based MTD*: MTD techniques have been modeled using genetic algorithms (GA). For the identification of a secured computer configuration with secure OS and application and higher diversity and time space, Crouse and Fulp[31] engaged genetic algorithms. They designed a computer configuration in such a way (i.e. chromosome) so that individual configuration can be considered as a trait. A high ranked configuration is elected on the basis of resilience of an attack. Particularly, they reflect vulnerability through an aged configuration by considering the aging aspect. Additionally, The authors further improved the quality of GA based MTD by considering the alteration of mutation and feedback from the security status and changed configuration according to the changed mutation. [74]

3) *Machine Learning based MTD*: Randomization HTML elements for web services to defend against bot attacks has been proposed by Vikram et. al [75]. The effect of the mentioned strategy heightened by the use of machine learning. The authors have evaluated their technique on the basis of measurement of the time overhead of page loading. ML based MTD technique helps to lower overhead and in the meantime defends attack effectively. Prediction based MTD technique has been proposed by Colbaugh and Glass [76] where the model lower the ability of an attacker to learn about defensive techniques. Author used the fact that attacker are eager to learn the defense mechanism and design an attack by reverse-engineering the learned defense technique. They elevated their approach by evaluating their proposed algorithm using a cybersecurity dataset. They have also considered a different approach to deal with identical problems. [77]

TABLE IV
DIFFERENCES BETWEEN MULTIPLE MTD APPROACHES

Approach	Pros	Cons
Game Theoretic (GT) MTD	Able to capture of e model interactions between players; identifying optimal solution under complex interactions between multiple parties	Issues associated with players bounded rationality (or irrationality) and/or misperception; high solution complexity
Genetic Algorithm (GA)-based MTD	Fast method to find good solutions; providing good understanding of how good are given solution is based on a fitness function	Scalability problem in a large number of generators; it may not find the optimal solution; uncertainty to find fitness
Machine learning (ML)-based MTD	Capturing and evolving attacker; high scalability and application applicability	Needing a large amount of data for training and modelling

E. Discussion on the MTD modeling

A number of trends have been found prevalent in this aforementioned MTD modeling based on MTD approaches.

1) *A large amount of game theory has been considered for MTD modeling*: Game theory based MTD poses a substantial amount of benefits in terms of flexibility. For that reason, game theory has been considered for modeling MTD techniques dominantly.

2) *GA based MTD can identify paradigmatic deployment*: For large sized problems genetic algorithms can offer optimal solutions. Low quality of scalability with higher intricacy for a huge scale environment is a considerable cons of GA based MTD. Additionally, relying on a unified entity to call for MTD decisions can not be granted for few distributed-driven systems.

3) *ML based MTD can stand against adversary's learning capability and elect optimum strategy*: On the premise of big amounts of data availability, ML can show higher and low-flaw performance in educating about attack patterns and electing the best solution. ML based MTD can be poised as expensive and less beneficial if small amount of data is present or the dynamicity of the dataset is comparatively low.

VI. KEY MTD TECHNIQUES AND THEIR IMPLEMENTATIONS

Recently MTD developed a system ,philosophy and manufacturing inquiry conclusion that have gone out. But their work disclosed only the idea and the protective plan of MTD. The experiment of various masonry and the movement exercise of MTD skills.

MTD Skill: Program –emerged Variation : By changing program -functionality to remove vulnerabilities or limit their disclosure .This technique is called a software manipulation. Accretion, section, and back in previous place of functionality are all as hand skill. Putting functions in a skillful manner taken in program, and each it remain in a protection region. Execution of the similar act in a view , and that creates the process of variation to throw out the obsession.

Length of time-emerged Variation: In the Length of time-emerged Variation there are many defenses strategy that abolish attacks. Location Space layout is a broadly paradigm of Length of time- emerged Variation skill . In security skill, it garnish the onset the positions of premier code and information that ground in systems, location distance including the memory residence.

Relation Variation: In Relation Variation we hide our internal information or communication protocols .By relation skills we send our message or any information with each other .If communication skills is good ,attackers doesn't bother our any information ,But if anything is going to be wrong that's time relation variation helps us to protect our information . It also published the mock information in publically that would be helpful to indentify the true news and information could secret remote occlusion.

Progressive discussion skill: The another stage is Progressive discussion skill .It uses to transfer features to pull up attacking process .There are many variety to use in this platform. virtual machine rotation and diversity are broadly used as Progressive discussion skill for system strong .In

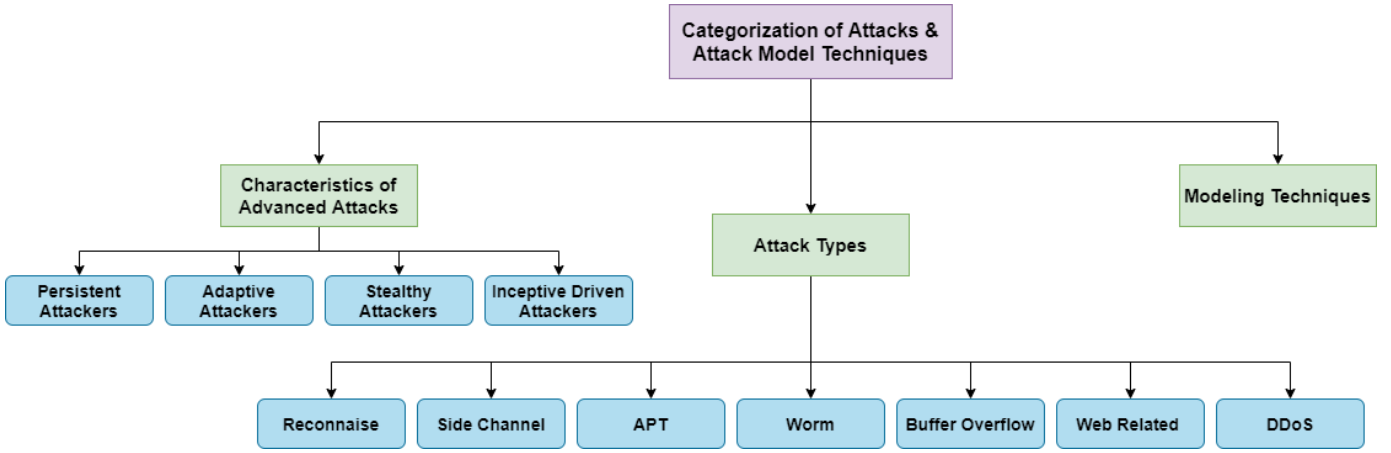


Fig. 4. Attack Categorization and Modelling Techniques

different platforms but one running applications can transfer and it would protect the whole system. At collector -target it investigate that we concentration on device target progressive discussion skill, which includes passage emerged skill, service in a network-pass skill, and oneself-mordant skill. Oneself-mordant skill can transfer running stage precedent except variation.

Mounting of MTD: The rare knowledge of system graph network culture are unavoidable, that creates network weakness. In many years ago ,many people doesn't know how to use internet .That's time many incident was happened for their rare knowledge.

VII. EVALUATION OF MTD SOLUTIONS

Since IOT has become popular in modern IT, MTD has also become a key technique. But the question is how to evaluate efficacy correctly. To quantitatively measure the efficacy of MTD strategies, researchers use security metrics in a few types of research. Taylor et al. [78], planned an assortment of metrics, including achievement, trustworthiness, efficiency, and privacy. Picek et al. [79] introduced a few metrics, for example, relative variety of the adequacy, normal holes of the general holes, and the normal length of the strolls, to surmise more data about the adequacy of MTD. Okhravi et al. [80] concerned the inclusion, unconventionality, timeliness and a metric to assess powerful development. Different researchers utilize the scientific model to assess the adequacy of MTD procedures. Here we will be mentioning some of them. By proposing a summed-up evaluation model using the Markov chain, Okhravi et al. [81] quantitatively evaluated the complex stage procedures as a protective portion. Maleki et al. [82] likewise utilized a discrete-time chain in Markov to demonstrate the co-operations among assailants and protectors in MTD games. Yet, it can't foresee the exhibition of an individual MTD solution. To tackle this constraint, Connell et al. [83] presented a quantitative, logical prototype, Continuous Time Markov Chains (CTMC), to evaluate the asset accessibility furthermore, execution of MTD. Now in this section, we

will try to introduce a shortlist of qualitative and quantitative metrics that can assist in assessing the efficacy of MTD solutions. [8]

A. Qualitative Evaluation

To evaluate the quality of a solution for MTD, we will look into two major features: security considerations and performance considerations. We will consider the nature of every individual defense solution under each of these subheadings and afterward, the general ensemble of constituent defence. FIG X shows a breakdown of this segment. So first we will explore the specifics of how the various works will evaluate the different qualitative features.

1) *Security Considerations:* We will evaluate this segment in 2 more sub-segments, Security of Individual Defenses and Security of the Ensemble [8]. These papers have considered the security of each individual defense in the entire ensemble by portraying them as part of the utility value of the defender [23], [26], [84], [85]. These works can concoct an intelligent development procedure dependent on thinking over known vulnerabilities (more explicitly, general vulnerabilities (CVE)) for which CVSS(Common Vulnerability Scoring Services) scores are promptly benefit capable. Different works that additionally just catch the security of person constituent framework setups are more space or issue explicitly in nature like in [86], authors pick an activity (to move a VM or continue running) in the wake of thinking about the protection hazards related with the existing vulnerable and weak state. The security hazard depends on the state of the existing situation, which progressively takes into account the form and number of attacks that were carried out in previous steps when the particular device was transmitted, and it is discussed in [30]. A few works also think of the longitudinal effects of a development methodology model the epistemic effects of development on an assailant's information over the entire network [67]. Now, the researches that mark just the security of an ensemble, instead of the security of individual activities, generally feature the security advantages of the MTD gatherings by contrasting them with the security privileges given through a latent

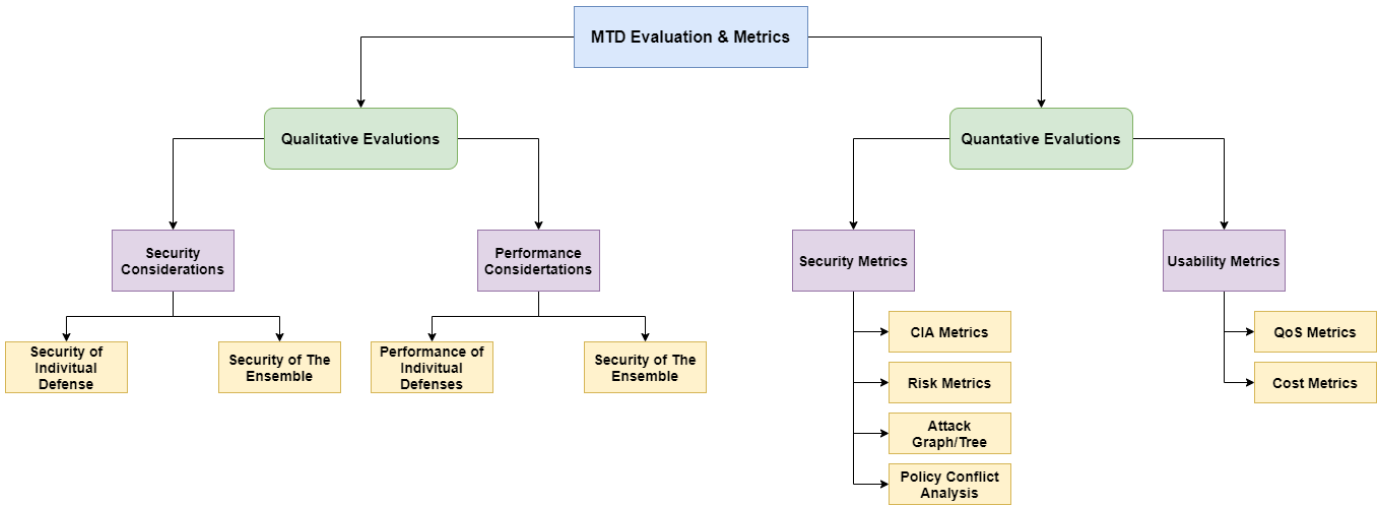


Fig. 5. Evaluation and Metrics of MTD Solutions

framework arrangement [87], [88]. Then again, researches that reflect the security measurements related to the assembly at demonstrating time take consideration at measurements that are like entropy or variety of the gathering [89], [90].

2) *Performance Considerations*: We will also evaluate this segment in 2 more sub-segments, Performance of individual defense actions and Performance of the ensemble [8]:

An enormous number of works consider problem-specific occurrences explicitly. In [91], authors perform investigations to think about their MTD in contrast to crossfire assaults and find that the normal time for parcel move starting with one point then onto the next increments on the grounds that choosing a specific way indiscriminately can be exceptionally problematic. In [84], the exhibition expenses of a honeypot arrangement, and this is the one that is considered as protection activity for this MTD solution ,expresses the figure and type regarding assets (or honey)required to build up each trustworthy honeypot that tricks an intruder.

Now, the works that measure just the execution of the ensemble attempt to catch every presentation expense that outcome due to the elements included are rearranging between the distinctive setup; however, don't consider at the exhibition impacts coming about on account of a specific terrible constituent design [70]. There are also different works which contrast a MTD guard with a static framework design by estimating ease of use measurements, for example, latency, accessibility to authentic clients,rather representing the exhibition effect of the ensemble [89], [90].

B. Quantitative Evaluation

To evaluate the quantity of an MTD solution we will look into two major features: Security Metrics, Usability Metrics. FIG Y shows a breakdown of this segment. Security metrics covers assault evaluation utilizing CVSS measurements Confidentiality, Integrity, and Availability (CIA) [83]. This metric additionally considers assault portrayal strategies (ARMs)—Attack Graphs and Attack Trees [92]. This metric also includes

Risk Metrics [16] and and important security analysis, Policy Conflict Analysis [93]. On the other hand Usability Metrics examines the analysis of MTD solutions from the features of QoS [94] (system bandwidth, delay), the effect on current purpose metrics, and the expense of applying the MTD defense solution recognized as cost metrics [31].

VIII. CONCLUSION & FUTURE RESEARCH DIRECTION

For the security and trust of a computer system, Moving Target Defense system is used which helps to detect several types of malicious behaviors in it. This approach for moving targets can be further extended towards investigating the classification process for any other high dimensional and imbalanced network addresses and their infrastructure. The detection of attacks is performed for malicious attacks and viruses here but other new types of attacks can also be detected and prevented by using this kind of defensive models in the perimeter layer. It is essential for researchers to improve their network based malicious system to be able to detect stealthy attacks in better manner, or combination with host based methods also be extended. And some malware and virus prevention techniques are also to be developed and added with the surveyed defense mechanism.

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